```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min(base + window_size - 1, total_frames - 1)
        print("\n----------------------------------------------")
        print(f"Sender Window : [Frame {base} - Frame {window_end}]")
        while next_seq < base + window_size and next_seq < total_frames:
            print(f"Node A --> Sending Frame {next_seq}")
            if next_seq == lost_frame:
                print(f"            (Frame {next_seq} LOST in transit!)")
            next_seq += 1
        if lost_frame is not None and base <= lost_frame < next_seq:
            print(f"Node B --> Frame {lost_frame} missing. Discarding out-of-order frames after it.")
            print(f"Node A --> Timeout! Resending window starting from Frame {lost_frame}.")
            base = lost_frame
            next_seq = lost_frame
            lost_frame = None
            continue
        for f in range(base, next_seq):
            print(f"Node B --> Received Frame {f}, sending ACK {f}")
        print(f"Node A --> ACK {next_seq - 1} received. Sliding window forward.")
        base = next_seq
    print("\n----------------------------------------------")
    print(f"All {total_frames} frames successfully transmitted from Node A to Node B.")
def main():
    print("=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===\n")
    total_frames = int(input("Enter total number of frames to send: "))
    window_size = int(input("Enter sender's window size: "))
    if total_frames <= 0 or window_size <= 0:
        print("Please enter positive values for total frames and window size.")
        return
    lost_frame = int(input(
        f"Enter frame number to simulate as LOST (0 to {total_frames - 1}, -1 for no loss): "
    ))
    if lost_frame == -1:
        lost_frame = None
    elif lost_frame < 0 or lost_frame >= total_frames:
        print("Invalid frame number.")
        return
    simulate(total_frames, window_size, lost_frame)
if __name__ == "__main__":
    main()
```

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```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send: 3
Enter sender's window size: 4
Enter frame number to simulate as LOST (0 to 2, -1 for no loss): 1

-----------------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
           (Frame 1 LOST in transit!)
Node A --> Sending Frame 2
Node B --> Frame 1 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 1.

-----------------------------------------------
Sender Window : [Frame 1 - Frame 2]
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

-----------------------------------------------
All 3 frames successfully transmitted from Node A to Node B.
```

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